

Brijlal Biyani Science College Amravati zoology

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Class; Bsc 3yr (sem 6th)

Group; BTZ

Seminar topic; Avery and co-workers experiments

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Avery, Macleod and Mc carty Experiment

- ◆ Avery, Macleod and Mc Carty repeated the experiment of bacterial transformation and identified nature of transforming substance.
- ◆ They separated the carbohydrate, protein, DNA like fractions of S-cell by variety of chemical and enzymatic methods.
- ❖ These fractions were added separately into nutrient medium containing R-cell and tested for their transforming ability as follows:
 1. When carbohydrates were added to nutrient medium containing R-cells, there was no transformation of R-cell to S-cells.
 2. When proteins were added, no transformation of R-cells to S-cell.
 3. When DNA estract was added, there was transformation of R-cell into S-cells.
 4. When DNA extract was added along with Dnase enzyme then there was no transformation. But when the extract was treated with protease the transformation ability was retained.

- ◈ In third set i.e., only in presence of DNA there is transformation of R-cells to S-cells.
- ◈ However, in fourth set DNA destroying enzyme was added so that transformation did not take place i.e., transforming ability of DNA is lost.
- ◈ Thus these experiments confirmed that transforming material located only on DNA and the genetic material is DNA. The transformation of genetic material from one cell to another by adding DNA extract of one cell in the culture is called transformation.

- Conclusion;-

As the transformation R-cell to S-cell took place when DNA extract was added, DNA is the genetic material.

e., transforming ability

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S strain
bacteria

Fractionation of cell free
extract into molecules

RNA

Protein

DNA

DNA+DNase

DNA+Protease

Molecules tested for transformation of R-strain cells

R

strain

R

strain

S

strain

R

strain

S

strain

entation of

Thankyou